

TraceFlow 1.0.0: Validation Pack

November 16, 2025

Version under test	
Engineer	
Tester	
Test Date	
Initial Report Generation Date	November 16, 2025
Result	
Approved Usage	

Manual tests carried out by:

Signature

Name

Job Title

Date

Validation pack result approved by:

Signature

Name

Job Title

Date

Contents

1	Requirements	3
1.1	Traceability Matrix	3
1.2	REQ-001: User authentication	3
1.2.1	Example LaTeX equation	3
1.2.2	Example inline code formatting	3
1.3	REQ-002: MRI dataset import	3
1.3.1	Example image (.png)	3
1.4	REQ-003: Image analysis pipeline	3
1.4.1	Example flow chart (using mermaid)	4
1.4.2	Example code block	4
1.5	REQ-004: Output data	4
1.6	REQ-005: Patient-study reconciliation	4
1.6.1	Example sequence diagram	4
2	Design - Functional Design Spec	5
2.1	Architecture overview	5
2.2	Identifier reconciliation engine (REQ-005, RISK-001, TEST-002)	5
2.2.1	UML activity view	5
2.3	Authentication and authorization (REQ-001, TEST-001)	5
2.4	Imaging pipeline (REQ-002, REQ-003, RISK-002)	6
2.5	Operational documents (General IUS)	6
3	Risk Register	7
3.1	RISK-001: Incorrect study-patient association	9
3.2	RISK-002: Undetected imaging artifacts	9
3.3	RISK-003: Audit trail corruption	9
4	Manual Acceptance Tests	10
4.1	TEST-001: User authentication	11
4.1.1	Test Steps:	11
4.1.2	Expected Result:	11
4.1.3	Test Outcome:	11
4.2	TEST-002: Automatic Playwright test	11
4.2.1	Test Steps:	11
4.2.2	Expected Result:	11
4.2.3	Test Outcome:	11
4.3	TEST-003: Automatic test	12
4.3.1	Test Steps:	12
4.3.2	Expected Result:	12
4.3.3	Test Outcome:	12
5	Installation & User Specification	14
5.1	Installation checklist	14
5.2	Operational monitoring	14
5.3	Traceability table	14
5.4	Sign-off	14

1 Requirements

These are some general notes on this set of requirements. Other content within subsections are for specific requirements.

1.1 Traceability Matrix

<i>Req ID</i> \ <i>Test ID</i>	TEST-001	TEST-002	TEST-003
REQ-001	✓		
REQ-002	✓		
REQ-003		✓	✓
REQ-004			
REQ-005	✓		

1.2 REQ-001: User authentication

Test ID: TEST-001

Users must be able to authenticate using their email address and a password.

1.2.1 Example LaTeX equation

The strength of the user's password should follow the entropy equation:

$$H = L \times \log_2(N)$$

Where H is the entropy, L is the password length, and N is the number of possible symbols.

1.2.2 Example inline code formatting

The ID of the user is, e.g., 1af345e6.

1.3 REQ-002: MRI dataset import

Test ID: TEST-001

The platform must support importing MRI datasets in DICOM format.

1.3.1 Example image (.png)



Figure 1: VoxelFlow Logo

1.4 REQ-003: Image analysis pipeline

Test ID: TEST-002 **Test ID:** TEST-003

- The platform should provide a Python API for building image analysis pipelines.
- The pipelines must be able to process MRI datasets in a compliant way.



Figure 2:

1.4.1 Example flow chart (using mermaid)

1.4.2 Example code block

```
def hello_world():
    print("Hello_world!")
```

1.5 REQ-004: Output data

The platform must be able to export the results of image analysis pipelines in a standard format, including the following parameters:

DICOM Series	Key Parameters	Typical Parameters	Map to Image Type
T1 VFA	TR, TE, FA	96x96x24 FA=2°,17°,32°	vfa
High-res pre-contrast	TR, TE, FA	512x512x92 FA=32°	high-res-pre
T1-weighted dynamic	TR, TE, FA, Temporal resolution	96x96x24 FA=17° Temporal resolution: 2s to 10s	dynamic-uncorrected
High-res post-contrast	TR, TE, FA	512x512x92 FA=32°	high-res-post

1.6 REQ-005: Patient-study reconciliation

Test ID: TEST-002

The ingestion service shall ensure that study metadata stays associated with the correct patient identifier to reduce **RISK-001** and **RISK-002**.

- Cross-validate the incoming HL7 and DICOM identifiers before persisting them.
- Flag mismatches for operator review and capture evidence via TEST-002.
- Provide a reconciliation webhook so downstream systems can resynchronize identifiers.

1.6.1 Example sequence diagram

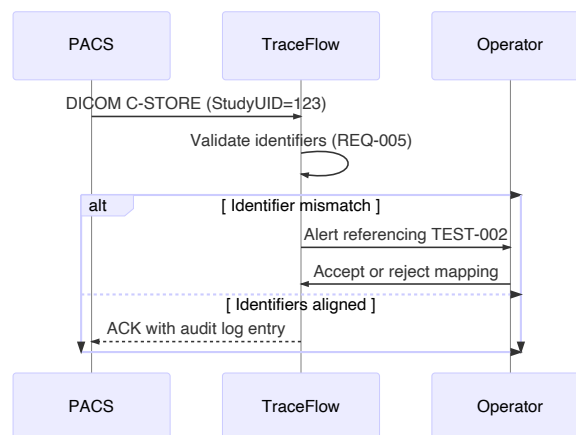


Figure 3:

2 Design - Functional Design Spec

This document provides a high-level architectural view of the TraceFlow demo and links each design decision to the requirements, tests, and clinical safety risks that drive them. Refer to **RISK-001** when assessing identifier safety controls described below.

2.1 Architecture overview

The system is split into ingestion, orchestration, and presentation tiers in order to satisfy **REQ-001** through **REQ-004** while maintaining a clean boundary for audit logging (**REQ-005**).

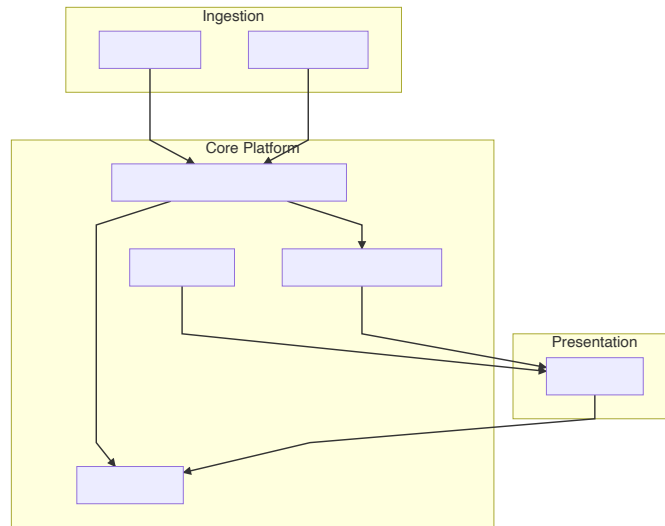


Figure 4:

2.2 Identifier reconciliation engine (REQ-005, RISK-001, TEST-002)

The reconciliation engine protects against the hazardous situation captured in **RISK-001** where patient and study identifiers might be mismatched. Design goals:

- Compare the DICOM PatientID and AccessionNumber with the HL7 metadata stream.
- If mismatches are detected, the transaction is quarantined and **TEST-002** exercises the operator override.
- Successful transactions emit structured audit events so **REQ-005** is met.

2.2.1 UML activity view

- Receive the incoming DICOM objects and perform schema checks.
- Compare HL7 metadata with extracted tags and branch depending on whether identifiers match.
- Persist successful studies and emit reconciliation metrics for **RISK-001**.
- Quarantine mismatches and require an operator acknowledgement that is captured by **TEST-002**.

2.3 Authentication and authorization (REQ-001, TEST-001)

Authentication relies on OpenID Connect so automated tests (**TEST-001**) can exercise both success and failure paths. Tokens carry the clinician's study permissions, and the middleware enforces them for every API endpoint.

2.4 Imaging pipeline (REQ-002, REQ-003, RISK-002)

The MRI import module normalizes DICOM before invoking the analysis pipelines described in **REQ-003**. The pipelines emit validation artifacts:

- DICOM specific logs for regulatory review.
- Image-derived measurements tied to their source requirement/test pairs.
- Alerts for saturation or out-of-range values, contributing to **RISK-002** mitigations.

2.5 Operational documents (General IUS)

Operational guidance for installations is maintained in `docs/ius.md`. That supplemental document links directly to **TEST-003** for continuous integration evidence and references **RISK-002** to remind the operator of the residual hazards.

3 Risk Register

TraceFlow libraries track software hazards alongside the requirements and tests that mitigate them.

Risk ID	Hazardous Situation	Harm	Cause	Severity (S)	Probability (P)	Risk Level	Risk Controls	Residual S	Residual P	Residual Risk	Linked Re-requirements	Linked Tests
RISK-001	Clinician views the wrong patient's images believing they are correct.	Misdiagnosis or inappropriate treatment.	Race condition during HL7/DICOM message processing leading to incorrect PatientID or Accession-Number.	High	Medium	High (12)	Identifier reconciliation service (REQ-005) exercised by TEST-002.	Medium	Low	Medium (6)		
Operator review plus automatic quarantine when mismatches occur.												
RISK-002	Poor quality MRI acquisitions flow through the analysis pipeline without alerts.	False clinical conclusions or delayed treatment.	Lack of automated QA gates between the import module (REQ-002) and analysis pipeline (REQ-003).	Medium	Medium	High (9)	Automated test matrix (TEST-003) and image QC hooks inside the pipeline.	Low	Low	Medium (4)		
Clinician-facing dashboard highlights residual warnings for manual acknowledgement.												
RISK-003	Investigators cannot re-construct prior runs because audit entries were dropped.	Compliance breach or inability to support safety investigations.	Misconfigured storage or missing webhook callbacks when persisting audit artifacts from REQ-005.	Medium	Low	Medium (6)	Continuous integration checks (TEST-003) and deployment checklist in docs/ius.md.	Low	Low	Medium (4)		
Acceptable provided long-term storage health checks pass weekly.												

3.1 RISK-001: Incorrect study-patient association

Hazardous Situation: Clinician views the wrong patient's images believing they are correct.

Harm: Misdiagnosis or inappropriate treatment.

Cause: Race condition during HL7/DICOM message processing leading to incorrect PatientID or AccessionNumber.

Severity: High

Probability: Medium

Controls: Identifier reconciliation service (**REQ-005**) exercised by **TEST-002**.

Residual Severity: Medium

Residual Probability: Low

Residual Risk: Operator review plus automatic quarantine when mismatches occur.

3.2 RISK-002: Undetected imaging artifacts

Hazardous Situation: Poor quality MRI acquisitions flow through the analysis pipeline without alerts.

Harm: False clinical conclusions or delayed treatment.

Cause: Lack of automated QA gates between the import module (**REQ-002**) and analysis pipeline (**REQ-003**).

Severity: Medium

Probability: Medium

Controls: Automated test matrix (**TEST-003**) and image QC hooks inside the pipeline.

Residual Severity: Low

Residual Probability: Low

Residual Risk: Clinician-facing dashboard highlights residual warnings for manual acknowledgement.

3.3 RISK-003: Audit trail corruption

Hazardous Situation: Investigators cannot reconstruct prior runs because audit entries were dropped.

Harm: Compliance breach or inability to support safety investigations.

Cause: Misconfigured storage or missing webhook callbacks when persisting audit artifacts from **REQ-005**.

Severity: Medium

Probability: Low

Controls: Continuous integration checks (**TEST-003**) and deployment checklist in docs/ius.md.

Residual Severity: Low

Residual Probability: Low

Residual Risk: Acceptable provided long-term storage health checks pass weekly.

4 Manual Acceptance Tests

These are some general notes on this set of test. Other content within subsections are for specific tests.

Tester	
Test Date	
Result	
Observations	

4.1 TEST-001: User authentication

Requirement ID: REQ-001 Requirement ID: REQ-002

4.1.1 Test Steps:

- Navigate to the login page.
- Enter a valid email address and password.
- Click the "Login" button.

4.1.2 Expected Result:

The user is logged in and redirected to the main dashboard.

4.1.3 Test Outcome:

Pass	Fail	Skip
Comments		

4.2 TEST-002: Automatic Playwright test

Requirement ID: REQ-003 Requirement ID: REQ-005

4.2.1 Test Steps:

- The `autoplaywright` block runs the Playwright sample located in the `playwright/` directory and captures the video, stdout/stderr, and a 3×3 key frame grid from the test execution.
- The scripted run intentionally drives a patient-identifier edit workflow to demonstrate the mitigation for **RISK-001** .

4.2.2 Expected Result:

The test should pass successfully.

4.2.3 Test Outcome:

Pass ☒	Fail ☐	Skip ☐
Playwright Test: 0001-test-counter		

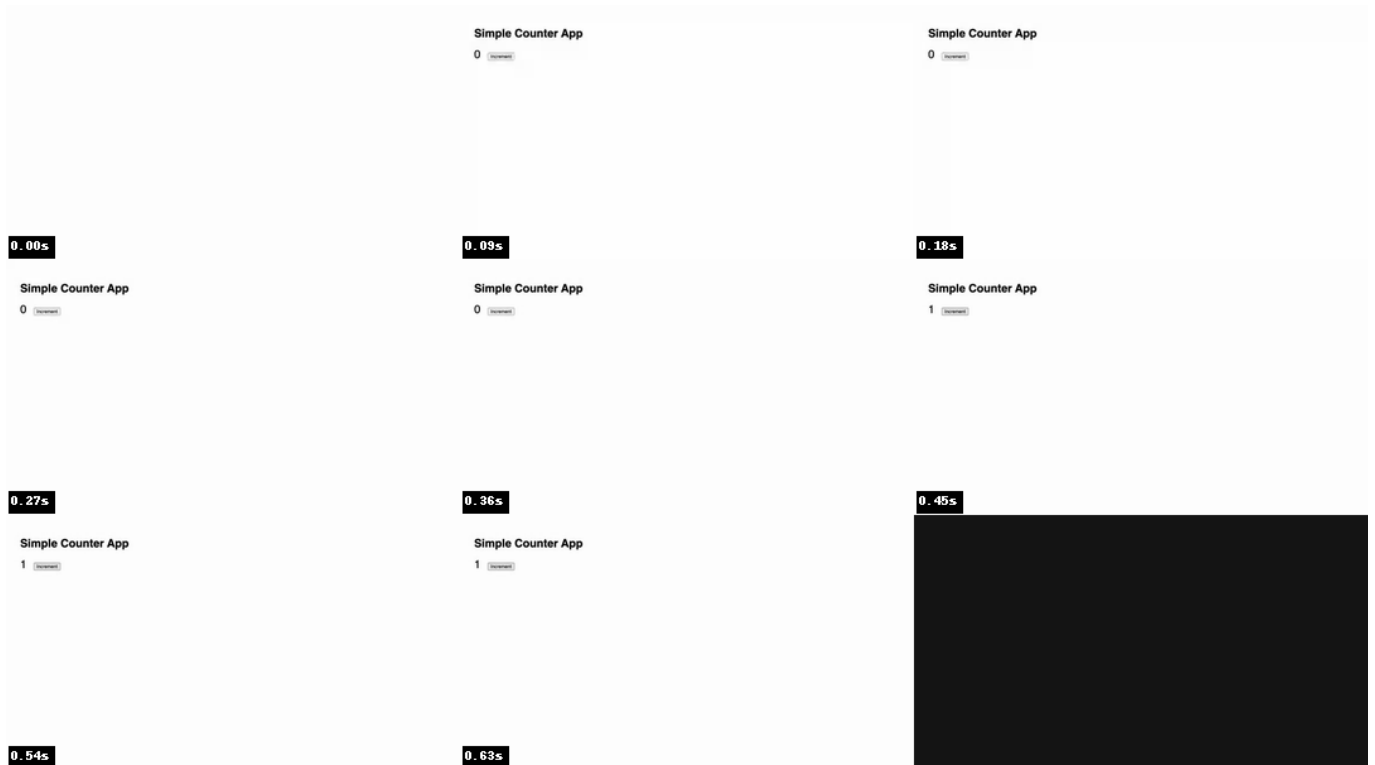


Figure 5: Key frames captured from 0001-test-counter

Running 1 test using 1 worker

```
[PASS] 1 counter.spec.js:5:1 > 0001-test-counter (891ms)
1 passed (1.5s)
```

4.3 TEST-003: Automatic test

Requirement ID: REQ-003

4.3.1 Test Steps:

- This autotest block demonstrates running a simple pytest command as part of the report.
- The quick-running suite doubles as a guardrail for **RISK-002** and **RISK-003** ; failures block deployments.

4.3.2 Expected Result:

The test should pass successfully.

4.3.3 Test Outcome:

Pass ☒ Fail ☐ Skip ☐

Playwright Test: 0001-test-counter

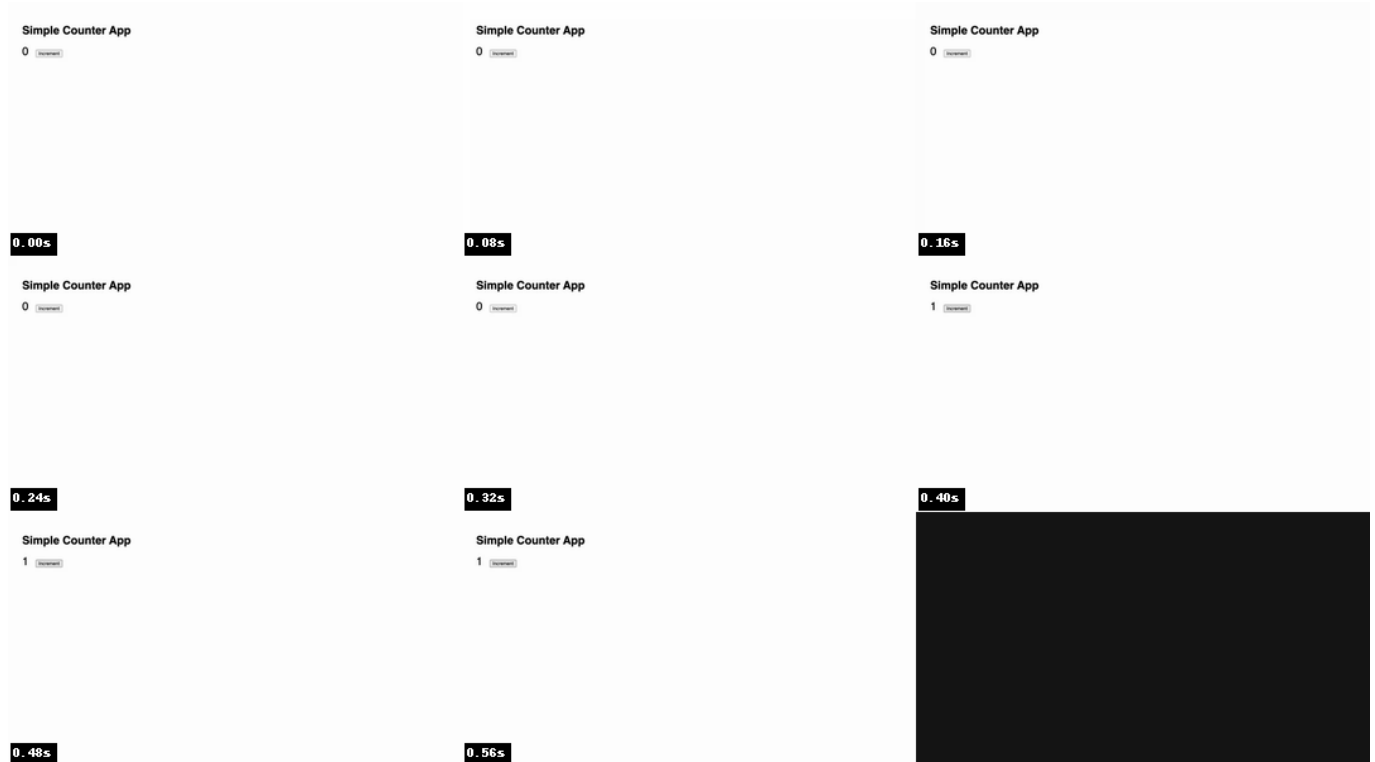


Figure 6: Key frames captured from 0001-test-counter

Running 1 test using 1 worker

[PASS] 1 counter.spec.js:5:1 > 0001-test-counter (763ms)

1 passed (1.3s)

Pass ☒ Fail ☐ Skip ☐

```
===== test session starts =====
platform darwin -- Python 3.11.13, pytest-9.0.1, pluggy-1.6.0 --
/Users/matthew.heaton/Work/SoftwareDevelopment/TraceFlow/.venv/bin/python3
cachedir: .pytest_cache
rootdir: /Users/matthew.heaton/Work/SoftwareDevelopment/TraceFlow
configfile: pyproject.toml
collecting ... collected 1 item

tests/test_hello_world.py::TestHelloWorld::test_zero Hello world!
PASSED

===== 1 passed in 0.00s =====
```

5 Installation & User Specification

This generic operational document demonstrates how TraceFlow can include arbitrary Markdown artifacts that link back to requirements, tests, and risks.

5.1 Installation checklist

- Deploy the ingestion stack and confirm that the `traceflow` service exposes `/healthz`.
- Verify that authentication (**REQ-001**) succeeds for a clinical account with the `clinician` role by following **TEST-001**.
- Enable the identifier reconciliation engine (**REQ-005**) and record the resulting audit entry.
- Capture Playwright evidence from **TEST-003** and attach it to the release ticket.

5.2 Operational monitoring

- Operators review the reconciler dashboard hourly to ensure the guardrails for **RISK-001** and **RISK-002** remain in place.
- When a mismatch occurs, follow the escalation steps documented in the clinical SOP and reference the impacted **REQ-005** controls.
- Residual risks shall be re-evaluated whenever a new modality is onboarded.

5.3 Traceability table

Step	Linked Requirement/Test	Linked Risk
Configure audit webhooks	REQ-005	RISK-001
Re-run integration suite after upgrade	TEST-003	RISK-002
Attach validation pack to submission	TEST-001 / TEST-002	RISK-003

5.4 Sign-off

The implementation owner confirms that installation and operational verification were executed per procedure and that the TraceFlow validation pack is archived with the evidence references listed above.